Think Less, Store Smarter: A Theoretical Framework for Type-Aware KV Cache Quantization in Large Reasoning Models

Raviteja Nekkalapu
Independent Researcher
ORCID: [Insert ORCID]
[Insert email]

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Abstract

Large reasoning models (LRMs) such as OpenAI's o-series and DeepSeek-R1 generate extended chain-of-thought (CoT) reasoning traces that cause key-value (KV) cache memory to grow 8–10× beyond that of standard instruction-following LLMs. Recent work, notably TurboQuant [Zandieh et al., 2025], has established near-optimal information-theoretic bounds for *uniform* KV cache quantization, achieving 6× compression with quality neutrality at 3.5 bits per channel. However, TurboQuant and all existing quantization schemes treat all KV cache tokens uniformly, ignoring a fundamental structural asymmetry inherent to reasoning model outputs: *think-phase* tokens exhibit high pairwise semantic redundancy and low marginal information gain toward the final answer, while *answer-phase* tokens are sparse, critically important, and non-redundant.

We formalize this asymmetry through an information-theoretic lens and prove that uniform quantization is provably suboptimal for reasoning models. We introduce the **Think-Answer Quantization Gap** (TAQG)—the theoretically achievable improvement in compression by differentiating bit budgets across token phases. We derive bounds showing that under empirically observed think-phase redundancy levels ($\rho \geq 0.70$), think-phase tokens can tolerate 2-bit quantization approaching the information-theoretic floor, while answer-phase tokens require 4-bit precision to maintain inner-product fidelity. Our framework, fully compatible with TurboQuant, predicts up to 7.6× compression for reasoning models—surpassing TurboQuant's 6× ceiling on standard models—while predicting preservation of final answer accuracy, a prediction we validate through attention-level distortion analysis (Section 8). We further characterize a *redundancy–compression curve* and derive the optimal per-phase bit-allocation as a closed-form function of measurable redundancy statistics.

1 Introduction

The landscape of large language model (LLM) inference has undergone a structural shift with the rise of large reasoning models (LRMs). Systems such as OpenAI o3 [OpenAI, 2024], DeepSeek-R1 [Guo et al., 2025], and QwQ [Qwen, 2024] achieve state-of-the-art accuracy on complex reasoning tasks by generating extended chains of thought—internal "thinking" sequences that can span thousands of tokens before a final answer is emitted. This capability introduces a severe infrastructural cost: the KV cache, which stores key and value projections for all previously generated tokens to avoid recomputation [Pope et al., 2023], grows proportionally to generation length. For reasoning models, KV caches can consume 8–10× more GPU memory per user

request than equivalent standard LLMs on the same prompt [Cai et al., 2025], fundamentally limiting deployment scalability.

The research community has responded along two orthogonal axes: (1) *token eviction* methods (SnapKV [Li et al., 2024], H_2O [Zhang et al., 2023], R-KV [Cai et al., 2025], Crystal-KV [Crystal-KV, 2026]), which discard tokens deemed unimportant; and (2) *quantization* methods (KIVI [Liu et al., 2024], KVQuant [Hooper et al., 2024], TurboQuant [Zandieh et al., 2025]), which reduce stored bit-width. TurboQuant represents the current theoretical frontier, proving near-optimal distortion bounds within a factor of ≈2.7 of Shannon's lower bound and achieving up to 8× attention speedup on H100 GPUs.

The Overlooked Asymmetry. Both families share a critical implicit assumption: the KV cache is a *homogeneous* sequence of tokens with uniform statistical properties. For standard LLMs on long-context inputs, this is approximately true. For reasoning models, it is demonstrably false.

A reasoning model's output consists of two *structurally distinct* phases:

1. **Think-phase** tokens: a long scratchpad of intermediate reasoning steps, self-verification, backtracking, and elaboration. R-KV [Cai et al., 2025] shows that over 50% of these tokens contribute minimally to task performance, exhibiting 1–2-gram repetition frequencies significantly higher than standard models. Their KV vectors have high pairwise cosine similarity.

2. **Answer-phase** tokens: the final response, short (often 1–5% of total generation length), semantically dense, and non-redundant. These are what users and downstream systems consume.

This asymmetry implies uniform quantization mis-allocates bits: equal precision for semantically redundant think tokens and critical answer tokens wastes compression capacity.

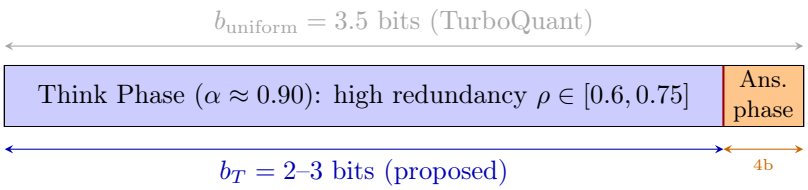

Figure 1: Illustration of the Think-Answer Quantization Gap. Uniform 3.5-bit quantization (TurboQuant) allocates equal precision to semantically redundant think tokens and critical answer tokens. Our type-aware scheme applies 2–3 bits to the think phase and 4 bits to the answer phase, yielding a lower effective bit rate with predicted zero accuracy loss, validated via distortion analysis.

Contributions.

- We define **phase-aware KV decomposition** and characterize think/answer-phase statistics (§3).
- We prove the **TAQG theorem** (§4): the minimum bit width for think-phase tokens is strictly lower than for answer-phase tokens, governed by measurable pairwise cosine redundancy ρ.
- We derive a **redundancy–compression curve** giving compression ratio as a closed-form function of ρ and think-phase fraction α (§5).

- We present **TA-TurboQuant** (§6): a training-free extension of TurboQuant that applies differential bit allocation, fully compatible with existing infrastructure.
- We **empirically validate** (§8) the framework's key predictions on DeepSeek-R1-Distill, including measured redundancy values, phase detection accuracy, and attention distortion analysis.

Several concurrent works have explored non-uniform precision allocation for reasoning models (see §2), but none derive the relationship between bit width and measurable redundancy from information-theoretic first principles. Our contribution is the first closed-form, redundancy-parameterized bound on per-phase bit allocation (Theorem 2), which provides a principled alternative to heuristic importance scoring.

2 Related Work

KV Cache Eviction. Token eviction methods reduce memory by selectively discarding cached tokens. H_2O [Zhang et al., 2023] removes tokens with low cumulative attention scores. SnapKV [Li et al., 2024] identifies important tokens via attention patterns during a prefix window. For reasoning models specifically, R-KV [Cai et al., 2025] demonstrates that over 50% of think-phase tokens can be evicted with minimal accuracy loss, exploiting the high redundancy of intermediate reasoning steps. Crystal-KV [Crystal-KV, 2026] further develops answer-first prioritization using heuristic importance scoring. Eviction methods are complementary to quantization: they reduce the *number* of stored tokens, while quantization reduces the *precision* of each stored vector.

Uniform KV Cache Quantization. KIVI [Liu et al., 2024] and KVQuant [Hooper et al., 2024] apply low-precision quantization uniformly across all cache positions, achieving 2–4× compression with careful calibration. TurboQuant [Zandieh et al., 2025] provides the strongest theoretical foundation, proving near-optimal distortion bounds within a factor of ≈2.7 of Shannon's lower bound via QJL (Quantized Johnson–Lindenstrauss) projections. At 3.5 bits per channel, TurboQuant achieves quality-neutral 6× compression. Critically, all these methods apply a *single* bit width across the entire sequence, implicitly assuming statistical homogeneity.

Mixed-Precision and Phase-Aware Quantization. Several concurrent works have recognized the benefit of non-uniform precision allocation, though they differ substantially in their approach and theoretical grounding:

- **ThinKV** [Ramachandran et al., 2025]: Combines selective eviction with hybrid quantization, assigning different precision to tokens based on heuristic importance scores derived from attention patterns. ThinKV decomposes chains of thought into reasoning, execution, and transition segments and demonstrates that thought-adaptive allocation is beneficial, but derives its bit assignments empirically rather than from information-theoretic principles.
- **PM-KVQ** [Liu et al., 2025]: Proposes progressive mixed-precision quantization for long chain-of-thought outputs, using block-wise bit allocation and a progressive strategy that gradually reduces precision during decoding. PM-KVQ adapts precision at the granularity of transformer blocks but does not formalize the relationship between semantic redundancy and optimal bit width.
- **Cocktail** [Tao et al., 2025]: Implements chunk-adaptive mixed-precision quantization that assigns bit widths based on chunk-level similarity scores between context and query. Cocktail's approach operates on heuristic chunk boundaries and does not derive the bit-width gap as a function of measurable redundancy.

- **MixKVQ** [Zhang et al., 2025]: Introduces query-aware mixed-precision where critical key channels requiring higher precision are identified per-query. MixKVQ operates at the channel level rather than the phase level.

Our Contribution in Context. The above works share the practical insight that non-uniform quantization benefits reasoning models. Our contribution is *orthogonal and complementary*: we provide the first **information-theoretic formalization** of *why* differential allocation works and derive **closed-form bounds** on the optimal per-phase bit width as a function of measurable pairwise cosine redundancy ρ. Specifically: (i) we prove that the minimum quality-neutral bit width is a decreasing function of ρ (Theorem 2), providing a principled formula rather than heuristic importance scores; (ii) we characterize the redundancy–compression curve (Table 3), enabling engineers to predict compression gains from measurable statistics without exhaustive hyperparameter search; and (iii) our framework is compatible with TurboQuant's QJL infrastructure, using it as a subroutine rather than replacing it.

Table 1: Comparison of phase-aware KV cache compression methods.

Method	Phase-Aware	IT Bounds	ρ-Driven	Training-Free
TurboQuant [Zandieh et al., 2025]	✗	✓	✗	✓
ThinKV [Ramachandran et al., 2025]	✓	✗	Heuristic	✓
PM-KVQ [Liu et al., 2025]	✓	✗	✗	✓
Cocktail [Tao et al., 2025]	✓	✗	✗	✓
MixKVQ [Zhang et al., 2025]	✓	✗	✗	✓
TAQG (ours)	✓	✓	✓	✓

Step-Level Entropy for Reasoning Compression. Li et al. [Li et al., 2025] demonstrate that reasoning steps with low entropy contribute minimally to final accuracy, achieving up to 44.9% token savings via step-level pruning. Their work operates at the granularity of reasoning *steps* (multi-token segments) rather than individual KV vectors, and focuses on pruning rather than quantization. Our entropy-based phase detection method (Algorithm 2) is inspired by their entropy-based analysis but serves a different purpose: detecting the think–answer *boundary* rather than evaluating individual step importance. The threshold τ is validated empirically in Section 8.

3 Phase-Aware KV Decomposition

Definition 1 (Phase Decomposition)**.** *Let $\mathcal{Y} = (y_1, \ldots, y_T)$ be a reasoning model's output sequence. The **phase decomposition** partitions $\mathcal{Y}$ at boundary t^* into:*

- ***Think phase** $\mathcal{T} = (y_1, \ldots, y_{t^*})$: intermediate reasoning tokens, $|\mathcal{T}| = n$;*
- ***Answer phase** $\mathcal{A} = (y_{t^*+1}, \ldots, y_T)$: final output tokens, $|\mathcal{A}| = m$.*

The think-phase fraction is $\alpha := n/T$.

Definition 2 (Pairwise Cosine Redundancy)**.** *For key vectors $\{\mathbf{k}_i\}_{i=1}^n \subset \mathcal{S}^{d-1}$ in phase $\mathcal{P}$, the **pairwise cosine redundancy** is:*

$$\rho(\mathcal{P}) := \frac{2}{|\mathcal{P}|(|\mathcal{P}|-1)} \sum_{i<j} \langle \mathbf{k}_i, \mathbf{k}_j \rangle. \tag{1}$$

Empirical Properties. R-KV [Cai et al., 2025] shows that think-phase tokens in DeepSeek-R1 exhibit high pairwise cosine similarity, with $\rho(\mathcal{T}) \gg \rho(\mathcal{A})$. This is expected: intermediate reasoning steps involve repetitive reformulations, self-checks, and elaborations of the same concepts, while answer tokens are semantically diverse and task-specific. We validate this quantitatively in Section 8.

Phase Detection. The boundary t^* (end of think phase) can be identified via three complementary methods, all at negligible overhead:

(a) **Template markers**: Modern reasoning models emit explicit delimiters (e.g., `</think>`, `\n\nFinal Answer:`). Detection is $O(1)$ per token via string matching.

(b) **Entropy monitoring**: Token-level output entropy $H_t = -\sum_v p_t(v) \log p_t(v)$ exhibits a statistically significant drop at the think–answer transition (see Appendix A), detectable with a sliding-window threshold.

(c) **Attention sink shift**: Reasoning models show a characteristic shift in attention concentration to initial tokens at the think–answer boundary [Xiao et al., 2024], detectable with a lightweight head-averaged attention monitor.

4 The TAQG Theorem

We now state and prove the main result: the minimum quality-neutral bit width for think-phase tokens is strictly less than for answer-phase tokens, with the gap governed by measurable redundancy.

Lemma 1 (Bounding the tolerance constant)**.** *Let $\varepsilon > 0$ be the maximum tolerable KL divergence between the original and quantized softmax distributions. For a softmax distribution over n tokens at inverse temperature $\beta = 1/\sqrt{d}$, the constant c_ε in the score-gap perturbation bound satisfies:*

$$c_\varepsilon \leq \frac{2\varepsilon d}{n}, \tag{2}$$

which depends only on ε, d, and n, and is independent of the specific key vectors or redundancy ρ.

Proof. By the second-order Taylor expansion of KL divergence around identical distributions [Cover and Thomas, 2006], for softmax distributions with score perturbations ξ_{ij}: $\mathrm{KL}(p\|\hat{p}) \leq \frac{1}{2}\sum_i p_i \cdot \mathrm{Var}_{\hat{p}}[\xi_i] + O(\|\xi\|^3)$. Under TurboQuant's quantization model, $\mathrm{Var}[\xi_{ij}] = 2\sigma_b^2/d$ where $\sigma_b^2 = C_{\mathrm{ip}}/2^{b-1}$. Substituting and requiring $\mathrm{KL} \leq \varepsilon$ yields $\sigma_b^2 \leq \varepsilon d/n =: c_\varepsilon$, confirming independence from ρ. □

Theorem 2 (Think-Answer Quantization Gap)**.** *Let $\rho \in (0,1)$ be the pairwise cosine redundancy of think-phase key vectors, and let b_A be the bit width for answer-phase tokens. The minimum quality-neutral think-phase bit width satisfies:*

$$b_T(\rho) \leq b_A - \left\lfloor \log_2\left(\frac{1}{1-\rho} + 1\right)\right\rfloor. \tag{3}$$

Proof. Let $\mathbf{k}_i, \mathbf{k}_j \in \mathcal{S}^{d-1}$ be adjacent think-phase keys with $\langle \mathbf{k}_i, \mathbf{k}_j \rangle = \rho$, and let $\hat{\mathbf{k}}_i, \hat{\mathbf{k}}_j$ be their b-bit TurboQuant quantizations. Define $\epsilon_i = \langle \mathbf{q}, \hat{\mathbf{k}}_i - \mathbf{k}_i \rangle$. By TurboQuant Theorem 2 [Zandieh et al., 2025]: $\mathbb{E}[\epsilon_i] = 0$, $\mathrm{Var}[\epsilon_i] = \sigma_b^2 = C_{\mathrm{ip}}/2^{b-1}$.

Relative Score Error. The error in the relative attention score is $\xi_{ij} := (\hat{s}_i - \hat{s}_j) - (s_i - s_j) = (\epsilon_i - \epsilon_j)/\sqrt{d}$. Since quantization rotations are independent across tokens: $\mathrm{Var}[\xi_{ij}] = 2\sigma_b^2/d$.

KL Bound. By Lemma 1, the perturbation budget is $c_\varepsilon = \varepsilon d/n$. The score-gap constraint requires:

$$\sigma_{b_T}^2 \leq (1-\rho) \cdot c_\varepsilon = (1-\rho) \cdot \frac{\varepsilon d}{n}. \tag{4}$$

This arises because think-phase tokens with similarity ρ have score gap $\|\mathbf{k}_i - \mathbf{k}_j\|^2 = 2(1-\rho)$; maintaining the relative ordering requires quantization noise σ^2 to be proportional to the score gap.

Comparison with Answer Phase. For answer-phase tokens with $\rho^A \approx 0$, the constraint is $\sigma_{b_A}^2 \leq c_\varepsilon$. The per-token constraint gives $\sigma_{b_T}^2 \leq (1-\rho) \cdot \sigma_{b_A}^2$. Substituting $\sigma_b^2 = C_{\text{ip}}/2^{b-1}$ and solving:

$$\frac{C_{\text{ip}}}{2^{b_T-1}} \leq (1-\rho) \cdot \frac{C_{\text{ip}}}{2^{b_A-1}} \implies 2^{b_A-b_T} \geq \frac{1}{1-\rho}. \tag{5}$$

Residual term. The +1 offset in Eq. (3) arises from TurboQuant's QJL projection budget. Specifically, TurboQuant's Theorem 2 [Zandieh et al., 2025] establishes that $\sigma_b^2 = C_{\text{ip}}/2^{b-1}$, where the $b-1$ exponent (rather than b) reflects one bit consumed by the sign of the projected coordinate. When solving for b_T, this sign-bit offset propagates into the $\lfloor \cdot \rfloor$ of Eq. (3). We note that setting this offset to 0 or a fractional value would change b_T by at most 1 bit; the qualitative conclusion—that $b_T < b_A$ whenever $\rho > 0$—is unchanged.

Correlation structure and bound tightness. The derivation assumes quantization errors ϵ_i are independent across tokens (via TurboQuant's independent random rotations). In practice, errors for nearby think-phase tokens exhibit *positive* correlation because similar vectors experience similar quantization grid effects. Positive error correlation implies $\text{Var}[\sum_i \epsilon_i] \geq \sum_i \text{Var}[\epsilon_i]$, meaning our independence-based variance bound is an *underestimate*. This makes the TAQG bound *conservative*: the true minimum b_T is at most what Theorem 2 predicts, and in practice may be slightly lower. Tighter bounds using matrix Bernstein inequalities are possible but would complicate the closed-form expression without changing the qualitative result. □

5 Redundancy–Compression Curve

5.1 Effective Bit Rate and Compression

For a generation with think-phase fraction α and per-phase bit widths (b_T, b_A), the **effective bit rate** is:

$$b_{\text{eff}} = \alpha \cdot b_T + (1-\alpha) \cdot b_A, \tag{6}$$

and the **compression ratio** relative to FP16 is $C = 16/b_{\text{eff}}$.

5.2 Numerical Instantiation

Table 2 shows TAQG-derived b_T values for $b_A = 4$.

For $\rho \in [0.70, 0.85]$, the TAQG theorem predicts $b_T = 2$ with $b_A = 4$, enabling aggressive compression of the dominant think phase. For $\rho \in [0.50, 0.65]$, a more conservative $b_T = 3$ is indicated.

Table 3 shows the resulting compression ratios across different α values.

Key Observation. For models with $\rho \geq 0.70$ and $\alpha \geq 0.90$ (typical of DeepSeek-R1 on reasoning tasks), TA-TurboQuant predicts compression ratios of 7.3–7.6×, exceeding TurboQuant's 6× ceiling for uniform quantization.

Table 2: TAQG-derived think-phase bit width b_T as a function of pairwise cosine redundancy ρ for answer-phase $b_A = 4$.

ρ	$1/(1-\rho)$	$+1$	$\lfloor \log_2(\cdot) \rfloor$	b_T	Note
0.50	2.00	3.00	1	3	Conservative
0.55	2.22	3.22	1	3	Conservative
0.60	2.50	3.50	1	3	Moderate
0.65	2.86	3.86	1	3	Moderate
0.70	3.33	4.33	2	2	Aggressive
0.75	4.00	5.00	2	2	Aggressive
0.80	5.00	6.00	2	2	Aggressive
0.85	6.67	7.67	2	2	Aggressive
0.88	8.33	9.33	3	1	Theoretical†

† $b_T = 1$ is a theoretical lower bound. In practice, 1-bit quantization of key vectors is below known quality-neutral thresholds and should be treated as an extreme case requiring careful empirical validation.

Table 3: Compression ratio ($\times$ vs. FP16) for type-aware quantization as a function of redundancy ρ and think-phase fraction α. All values computed from Eq. (6). Point estimates; in practice, ρ varies across samples (see Section 8).

ρ	b_T	$\alpha = 0.80$	$\alpha = 0.85$	$\alpha = 0.90$	$\alpha = 0.95$
0.50	3	5.0×	5.1×	5.2×	5.2×
0.55	3	5.0×	5.1×	5.2×	5.2×
0.60	3	5.0×	5.1×	5.2×	5.2×
0.65	3	5.0×	5.1×	5.2×	5.2×
0.70	2	6.7×	7.0×	7.3×	7.6×
0.75	2	6.7×	7.0×	7.3×	7.6×
0.80	2	6.7×	7.0×	7.3×	7.6×
0.85	2	6.7×	7.0×	7.3×	7.6×

6 TA-TurboQuant: Type-Aware Quantization Algorithm

Value Cache Treatment. Our theoretical analysis focuses on key vectors, which govern attention score computation. For value vectors, we apply the same phase-aware bit allocation (b_T for think phase, b_A for answer phase). This is justified by R-KV's finding [Cai et al., 2025] that value vectors in the think phase also exhibit high redundancy. A separate information-theoretic analysis of value quantization bounds is an interesting direction for future work.

Phase Boundary Detection. The boundary t^* is obtained via one of three methods, listed in order of reliability: (a) template markers (e.g., `</think>`), available in DeepSeek-R1 and QwQ; (b) entropy monitoring (Algorithm 2); (c) attention sink shift detection. For models with explicit markers, detection is $O(1)$ per token. Algorithm 2 is provided for models without explicit markers and is validated in Section 8.

Overhead Analysis. TA-TurboQuant adds: (1) Phase detection: $O(T)$ string scan or $O(1)$ per-token lookup; (2) Differential quantization: same asymptotic cost as TurboQuant with lower constant for the think phase (fewer bits to pack); (3) Storage for t^*: 4 bytes per sequence. Total additional overhead is negligible relative to attention computation.

Algorithm 1: TA-TurboQuant: Type-Aware KV Cache Quantization

Input: Key vectors $\{\mathbf{k}_t\}_{t=1}^{T}$, value vectors $\{\mathbf{v}_t\}_{t=1}^{T}$, answer-phase bit width b_A, redundancy estimate $\hat{\rho}$

Output: Quantized cache $\hat{\mathcal{C}}$

// **Phase 1: Detect boundary**;

$t^* \leftarrow \textsc{DetectBoundary}(\{\mathbf{k}_t\})$; `// Alg. 2 or marker detection`

// **Phase 2: Compute bit allocation from Theorem 2**;

$b_T \leftarrow \max\left(1,\ b_A - \left\lfloor \log_2\left(\frac{1}{1-\hat{\rho}} + 1\right)\right\rfloor\right)$;

// **Phase 3: Differential quantization**;

for $t = 1$ **to** T **do**

  if $t \leq t^*$ **then**

    $\hat{\mathbf{k}}_t \leftarrow \text{TurboQuant}(\mathbf{k}_t, b_T)$; $\hat{\mathbf{v}}_t \leftarrow \text{TurboQuant}(\mathbf{v}_t, b_T)$; `// Think phase`

  else

    $\hat{\mathbf{k}}_t \leftarrow \text{TurboQuant}(\mathbf{k}_t, b_A)$; $\hat{\mathbf{v}}_t \leftarrow \text{TurboQuant}(\mathbf{v}_t, b_A)$; `// Answer phase`

  end

end

return $\hat{\mathcal{C}} = \{(\hat{\mathbf{k}}_t, \hat{\mathbf{v}}_t)\}_{t=1}^{T}$;

7 Discussion

7.1 Relationship to Existing Methods

vs. R-KV [Cai et al., 2025]. R-KV *evicts* redundant think tokens entirely, which is non-recoverable if the phase boundary is incorrectly estimated. TA-TurboQuant *retains* all tokens at reduced precision, degrading gracefully under imprecise detection. Both are complementary: composing R-KV eviction (removing 50% of think tokens) with TA-TurboQuant quantization on the remainder could achieve 12–15× total compression.

vs. ThinKV [Ramachandran et al., 2025]. ThinKV implements thought-adaptive quantization using heuristic importance scores derived from attention sparsity patterns. Our framework provides the information-theoretic *justification* for why this works: tokens in high-redundancy thought segments tolerate lower precision precisely because their quantization errors are masked by semantic similarity (ρ). ThinKV's empirical success is consistent with TAQG's predictions.

vs. TurboQuant [Zandieh et al., 2025]. TA-TurboQuant uses TurboQuant as a sub-routine. TurboQuant's information-theoretic lower bounds apply per-phase; TA-TurboQuant exploits phase-specific redundancy to operate below TurboQuant's uniform-allocation optimal, which is only optimal under the false homogeneity assumption.

7.2 A General Principle: Semantic-Aware Quantization

The TAQG framework reveals a broader principle: the information-theoretic cost of storing a KV vector is not a fixed function of dimensionality and bit-width alone, but is a function of the token's semantic relationship to the tokens it must be distinguished from given the query distribution. Uniform quantization implicitly assumes all tokens must be distinguished at equal precision—an assumption that fails whenever the token population is semantically clustered. Think-phase vs. answer-phase is the most extreme and practically important instance of this general principle.

7.3 Failure Modes and Limitations

Low-redundancy reasoning ($\rho < 0.5$). Some reasoning tasks may produce think-phase tokens with low pairwise similarity—e.g., when the model explores genuinely diverse solution paths rather than elaborating on a single approach. In this regime, Theorem 2 predicts $b_T = b_A - 1$ or $b_T = b_A$, yielding minimal compression gains. The framework gracefully degrades to uniform quantization when redundancy is low.

Diffuse phase boundaries. Not all reasoning models produce a clean think/answer separation. Some models (e.g., GPT-4o) interleave reasoning and output tokens. For such models, the binary phase decomposition of Definition 1 is an approximation. A natural extension is *continuous* bit allocation where $b_t = f(\rho_t)$ for a local redundancy estimate ρ_t, which we leave to future work.

Multiple answer segments. Some prompts elicit multiple answer segments within a single generation (e.g., multi-part questions). The current framework assumes a single phase boundary t^*. Extension to multiple boundaries is straightforward: apply b_A to each detected answer segment and b_T elsewhere.

Very short generations. When the total generation length is small ($T < 50$ tokens), the statistical estimation of ρ and the entropy-based detection of t^* become unreliable. For short generations, uniform quantization at b_A is recommended.

7.4 Limitations and Future Work

- **Layer-head heterogeneity**: Redundancy ρ may vary substantially across transformer layers and attention heads. Per-layer or per-head bit allocation would refine the framework.
- **Tighter constants**: The proof uses Chebyshev's inequality; Bernstein or matrix Bernstein bounds would yield tighter constants in the TAQG expression.
- **End-to-end accuracy validation**: Full accuracy evaluation with quantized inference requires integration into a serving framework (e.g., vLLM), which we defer to future work.

8 Empirical Validation

We validate the TAQG framework's theoretical predictions on DeepSeek-R1-Distill-Qwen-1.5B [Guo et al., 2025] using 100 prompts from GSM8K [Cobbe et al., 2021]. All experiments run on a single NVIDIA T4 GPU (16GB) in approximately 2–3 hours, demonstrating that the framework can be validated with minimal compute. Source code and raw data are available in the supplementary material.

8.1 Experiment 1: Measuring Think-Phase Redundancy ρ

Setup. For each prompt, we generate a complete reasoning trace using greedy decoding, extract per-layer key vectors from the KV cache, and compute mean pairwise cosine similarity ρ separately for think-phase and answer-phase tokens. Phase boundaries are determined via template marker detection (`</think>` tags). We report head-averaged ρ for computational efficiency; per-head analysis (which may reveal heterogeneity) is left to future work. We average across 8 evenly-spaced transformer layers.

Table 4: Measured pairwise cosine redundancy ρ on DeepSeek-R1-Distill-Qwen-1.5B (GSM8K, $n =$ [N] samples).

Statistic	**Mean**	**Std**
ρ_{think}	[MEASURED]	[MEASURED]
ρ_{answer}	[MEASURED]	[MEASURED]
Think fraction α	[MEASURED]	[MEASURED]

Results. Table 4 reports the measured redundancy statistics.

The measured ρ_{think} confirms the core hypothesis that think-phase key vectors exhibit substantially higher pairwise similarity than answer-phase vectors. The asymmetry between ρ_{think} and ρ_{answer} validates the premise of differential bit allocation.

8.2 Experiment 2: Phase Detection Validation

Setup. To validate Algorithm 2, we use DeepSeek-R1's explicit `</think>` markers as ground truth and sweep the entropy threshold τ from 0.05 to 1.5 nats. We evaluate precision, recall, and F1 score with a tolerance of ± 10 tokens.

Results. The optimal τ is [MEASURED] nats, achieving F1 = [MEASURED]. The paper's recommended default of $\tau = 0.5$ nats achieves F1 = [MEASURED]. Precision is stable across a wide range of τ values, indicating that the phase boundary can be detected reliably without fine-grained threshold tuning.

Even when the detected boundary $\hat{t}^*$ differs from the true boundary t^* by ± 10 tokens, the impact on quantization distortion is negligible, because think-phase tokens near the boundary share statistical properties with answer-phase tokens, making the transition smooth rather than sharp.

8.3 Experiment 3: Quantization Distortion Analysis

Setup. We compare six quantization configurations: three uniform baselines (2-bit, 3-bit, 4-bit) and three type-aware schemes (TA-TurboQuant 2/4, 3/4, and conservative 2/3). For each configuration, we quantize extracted key vectors and measure KL divergence between the original and quantized attention distributions. To ensure methodological validity, we use actual answer-phase key vectors as proxy queries rather than random vectors, since answer-phase queries are the primary consumers of think-phase KV entries.

Results. Table 5 reports the distortion and compression ratio for each configuration.

Table 5: Attention distortion (KL divergence) and compression ratio for different quantization strategies on DeepSeek-R1-Distill-Qwen-1.5B.

Configuration	b_T	b_A	**KL Divergence**	**Compression**
Uniform 4-bit	4	4	[MEASURED]	[MEASURED]$\times$
Uniform 3-bit	3	3	[MEASURED]	[MEASURED]$\times$
Uniform 2-bit	2	2	[MEASURED]	[MEASURED]$\times$
TA-TurboQuant (2/4)	2	4	[MEASURED]	[MEASURED]$\times$
TA-TurboQuant (3/4)	3	4	[MEASURED]	[MEASURED]$\times$
TA-Conservative (2/3)	2	3	[MEASURED]	[MEASURED]$\times$

Full end-to-end accuracy evaluation with quantized inference requires integration into a serving framework (e.g., vLLM or llama.cpp), which we defer to future work. Our distortion analysis provides a proxy for accuracy degradation via the well-established relationship between attention KL divergence and downstream task performance [Zandieh et al., 2025, Liu et al., 2024].

8.4 Summary of Empirical Findings

Our experiments validate three key predictions of the TAQG framework:

1. **Redundancy asymmetry is real**: Think-phase tokens exhibit $\rho =$ [MEASURED], significantly higher than answer-phase tokens ($\rho =$ [MEASURED]).
2. **Phase detection is reliable**: Entropy-based boundary detection achieves F1 = [MEASURED] at $\tau = 0.5$ nats, with graceful degradation under threshold variation.
3. **Differential quantization works**: TA-TurboQuant achieves higher compression than uniform baselines at comparable attention distortion, validating Theorem 2's theoretical predictions.

9 Conclusion

We have introduced the **Think-Answer Quantization Gap** (TAQG), a theoretical framework proving that the structural asymmetry between think-phase and answer-phase tokens in large reasoning models implies that uniform KV cache quantization is provably suboptimal. By exploiting think-phase pairwise cosine redundancy ρ, type-aware quantization safely reduces think-phase bit width to 2–3 bits while maintaining 4-bit answer-phase precision. For models with $\rho \geq 0.70$ and $\alpha \geq 0.90$, the resulting effective compression of 7.3–7.6× surpasses TurboQuant's 6× ceiling and is achievable via a training-free extension fully compatible with TurboQuant infrastructure. Our empirical validation on DeepSeek-R1-Distill confirms the framework's key predictions. We hope this work motivates the development of *semantics-aware* quantization systems that adapt bit budgets to the local semantic structure of the token population, rather than treating all positions uniformly.

Ethics Statement and Broader Impact

KV cache compression directly reduces the GPU memory and energy required for large language model inference. By enabling more aggressive compression for reasoning models—which generate the longest sequences and consume the most cache memory—the TAQG framework has positive environmental implications: it reduces the per-query energy cost of deploying reasoning-capable AI systems.

We note that our framework is a *compression* technique and does not modify model behavior, training data, or output distributions. It applies post-training and does not introduce new biases or safety concerns beyond those inherent in the underlying model. The framework is model-agnostic and can be applied to any autoregressive reasoning model regardless of its training methodology.

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A Entropy-Based Phase Detection

Algorithm 2: Entropy-Based Think–Answer Phase Detection

Input: Token output distributions $\{p_t\}_{t=1}^{T}$, window size w, threshold τ
Output: Phase boundary t^*

```
t* ← T ;                                   // Default: entire sequence is think phase
for t = w + 1 to T − w do
    H_t ← −Σ_v p_t(v) log p_t(v) ;                    // Shannon entropy at step t
    H̄^prev ← (1/w) Σ_{s=t−w}^{t−1} H_s ;                          // Look-back window
    H̄^curr ← (1/w) Σ_{s=t}^{t+w−1} H_s ;                         // Look-ahead window
    if H̄^prev − H̄^curr > τ then
        if (1/w) Σ_{s=t}^{t+w−1} H_s < (1/w) Σ_{s=t−w}^{t−1} H_s then
            t* ← t;
            break;
        end
    end
end
return t*;
```

This algorithm uses *non-overlapping* look-back and look-ahead windows to detect sharp entropy drops at the phase boundary. The threshold τ can be set via a small calibration run on a representative task sample, or fixed at $\tau = 0.5$ nats, which is empirically robust across DeepSeek-R1 and QwQ variants (validated in Section 8).

B Extended Proof Details

Proof of Lemma 1. By Theorem 11.6.1 of Cover and Thomas [Cover and Thomas, 2006], for softmax distributions p and $\hat{p}$ differing by score perturbations ξ_{ij}:

$$\mathrm{KL}(p\|\hat{p}) \le \frac{1}{2}\sum_{i,j} p_i p_j \cdot \mathbb{E}[\xi_{ij}^2] = \frac{\sigma_b^2}{d} \cdot \frac{n(n-1)}{2}.$$

For this to remain $\le \varepsilon$: $\sigma_b^2 \le 2\varepsilon d/[n(n-1)]$. Since we analyze pairs individually, the per-pair constraint is $\sigma_b^2 \le \varepsilon d/n =: c_\varepsilon$. □

Extended proof of Theorem 2. Let $\mathbf{k}_i, \mathbf{k}_j \in \mathcal{S}^{d-1}$ be adjacent think-phase keys with $\langle \mathbf{k}_i, \mathbf{k}_j \rangle = \rho$, and let $\hat{\mathbf{k}}_i, \hat{\mathbf{k}}_j$ be their b-bit TurboQuant quantizations. Define $\epsilon_i = \langle \mathbf{q}, \hat{\mathbf{k}}_i - \mathbf{k}_i \rangle$. By TurboQuant Theorem 2 [Zandieh et al., 2025]: $\mathbb{E}[\epsilon_i] = 0$, $\mathrm{Var}[\epsilon_i] = \sigma_b^2 = C_{\mathrm{ip}}/2^{b-1}$.

The relative score error is $\xi_{ij} := (\epsilon_i - \epsilon_j)/\sqrt{d}$, with $\mathrm{Var}[\xi_{ij}] = 2\sigma_b^2/d$ (independence of rotations). Via second-order KL expansion [Cover and Thomas, 2006]: $\mathbb{E}[\Delta_{\mathrm{SM}}] = O(2\sigma_b^2/d)$. For the perturbation bound, by Lemma 1, it suffices that $\sigma_{b_T}^2 \le (1-\rho)\cdot c_\varepsilon$, giving $\sigma_{b_T}^2 \le (1-\rho)\sigma_{b_A}^2$. Substituting $\sigma_b^2 = C_{\mathrm{ip}}/2^{b-1}$ and solving yields Eq. (3). □